Who's to Blame for Survey Instability:
Respondents with Nonexistent Preferences or Researchers with Flawed Measures?[1]

Gary King[2]

Institute for Quantitative Social Science
Harvard University

Society for Political Methodology, 7/18/2026

[1]Slides, paper, data: GaryKing.org/instability
[2]Joint work with Libby Jenke, University of Houston

A Quiz (for you, as a survey researcher)

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- Ask: any nontrivial, binary choice survey question

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Coin Flip

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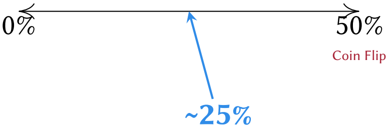

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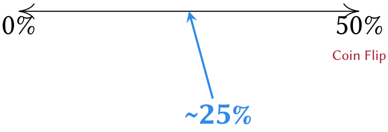

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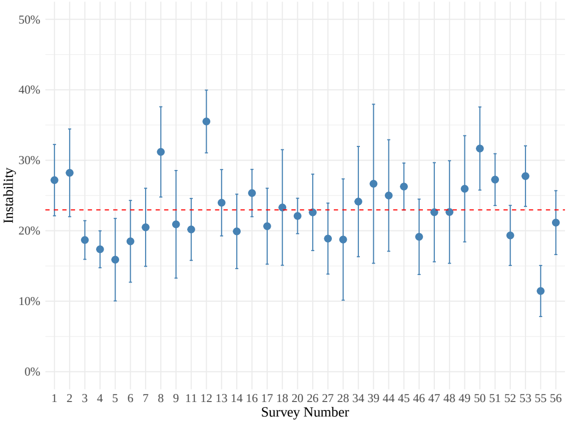

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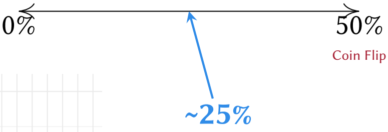

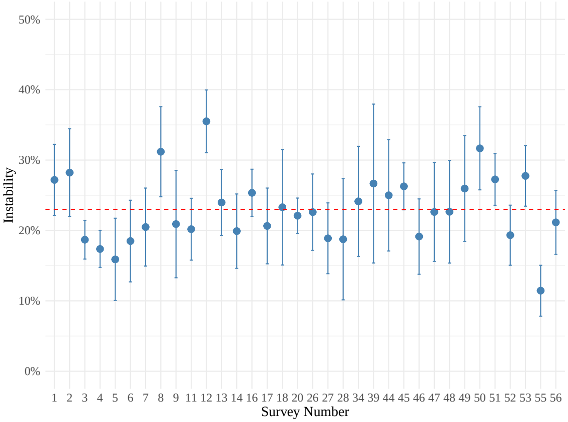

- Unchanged for 75 years
- Why trust any survey analysis?

Research Strategy

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on

A Binary Choice Conjoint Question

(Ono and Burden, 2018)

Please carefully review the two potential candidates running for election to the U.S. House of Representatives, detailed below.

	Candidate 0	**Candidate 1**
Race/Ethnicity	Hispanic	Asian American
Age	52	60
Favorability rating among the public	70%	34%
Position on immigrants	Favors giving citizenship or guest worker status to undocumented immigrants	Opposes giving citizenship or guest worker status to undocumented immigrants
Party affiliation	Republican Party	Democratic Party
Position on abortion	Abortion is not a private matter (pro-life)	Abortion is a private matter (pro-choice)
Position on government deficit	Wants to reduce the deficit through tax increase	Wants to reduce the deficit through tax increase
Salient personal characteristics	Really cares about people like you	Really cares about people like you
Position on national security	Wants to cut military budget and keep the U.S. out of war	Wants to maintain strong defense and increase U.S. influence
Gender	Female	Female
Policy area of expertise	Education	Foreign policy
Family	Single (divorced)	Married (no child)
Experience in public office	12 years	4 years

If you had to choose between them, which of these candidates would you vote to be a member of the U.S. House of Representatives?

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    - Reporting: every failed hypothesis & mistake

What We Know

Survey Instability Explanations

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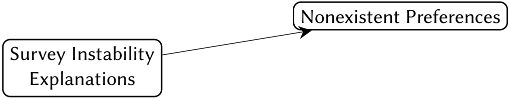
Nonexistent Preferences
Survey Instability
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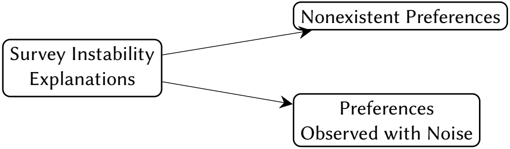
Survey Instability Explanations
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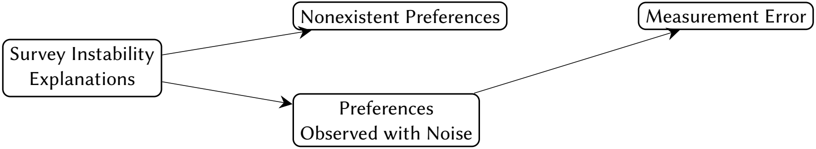
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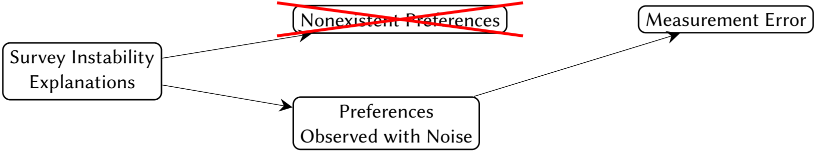
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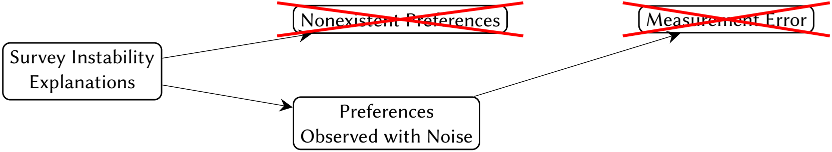
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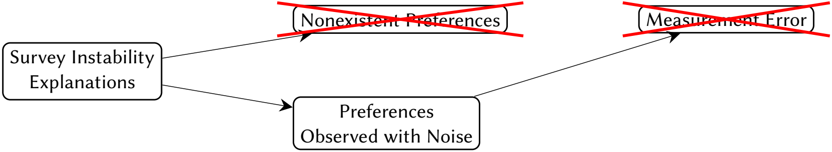

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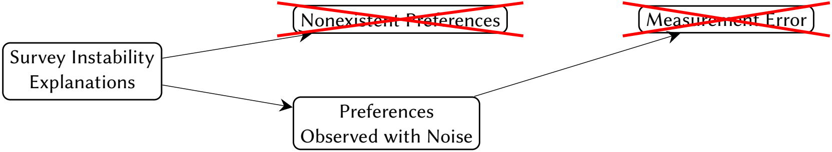

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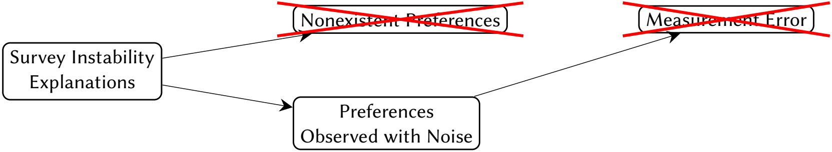

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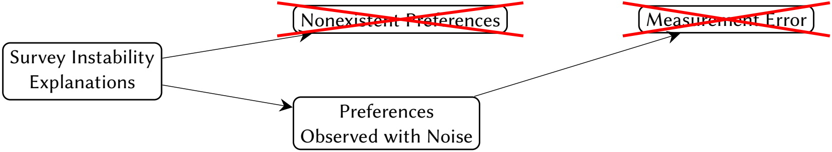

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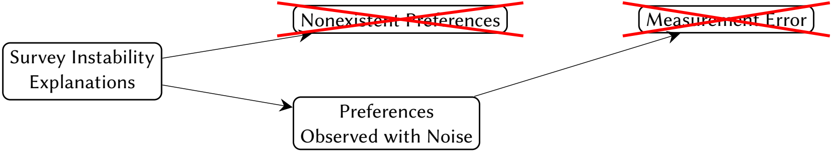

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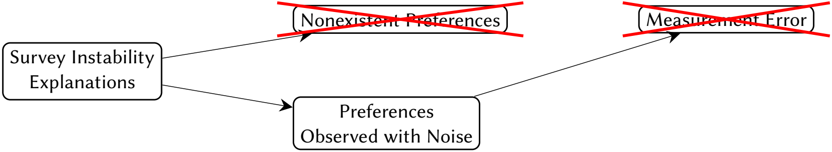

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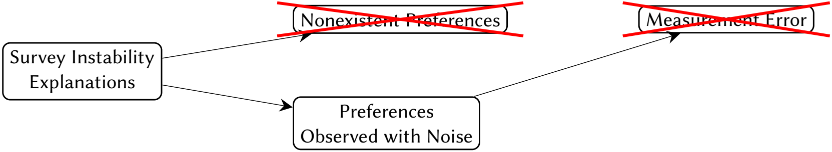

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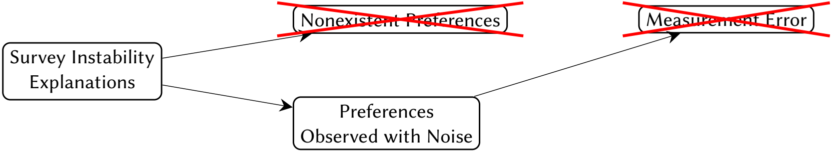

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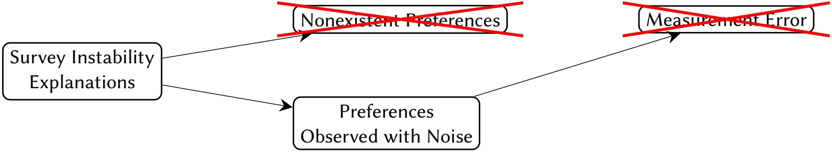

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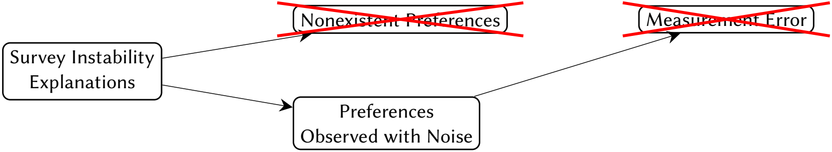

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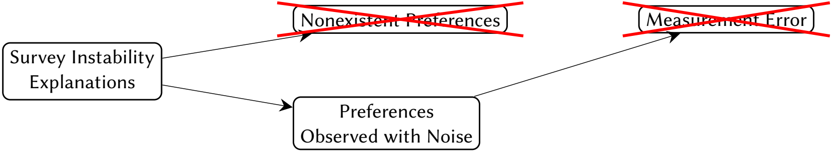

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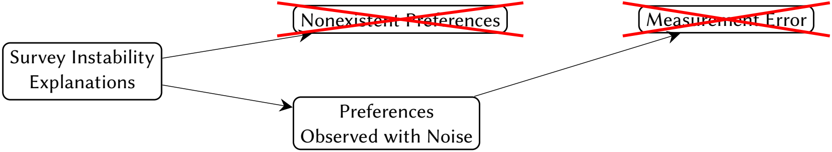

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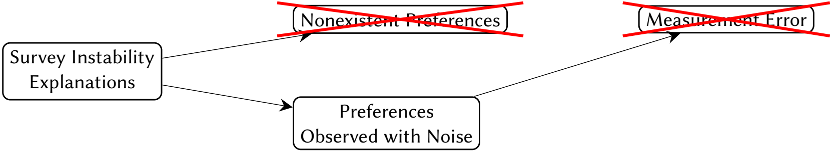
- Measurement Error
    - The Literature: must be M.E., because people have preferences!
    - Survey improvements:
        - affect <u>both</u> questions
        - reduce bias, not instability

What We Know



- ~~Nonexistent Preferences~~
  - ~all evidence contradicts Converse
  - Survey findings: numerous, strong, replicable
  - Persuasion attempts: usually fail
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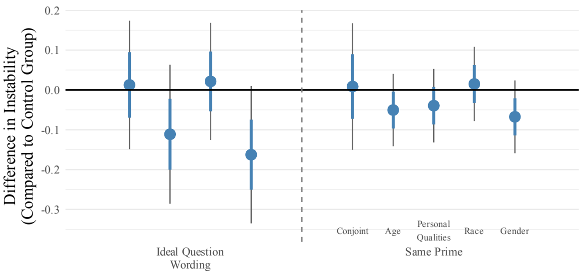
Survey Instability
Expla

Reducing Bias Doesn't Reduce Instability



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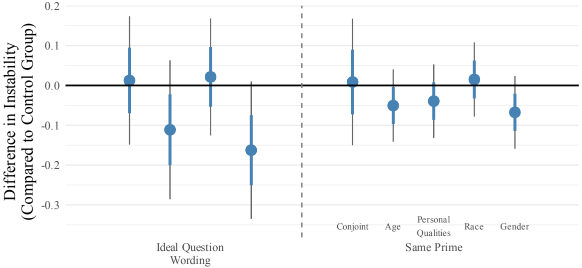

- Left: 2008 ANES question wording changes

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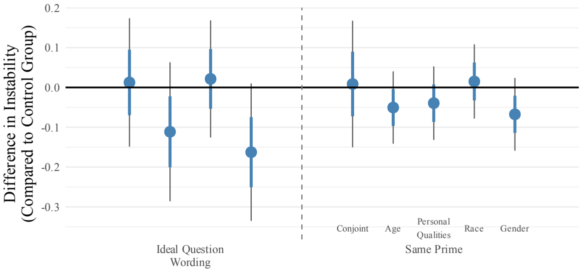

- Left: 2008 ANES question wording changes
- Right: different (priming) "considerations"

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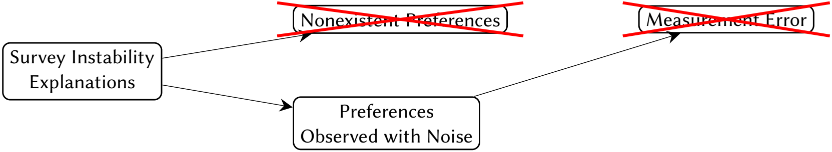

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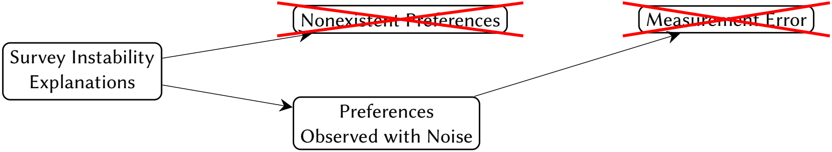

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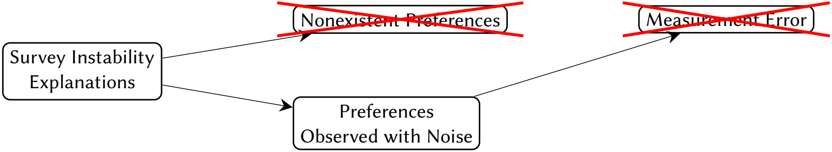

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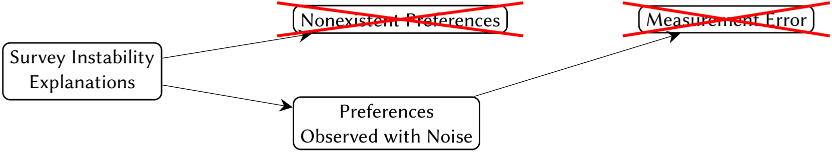
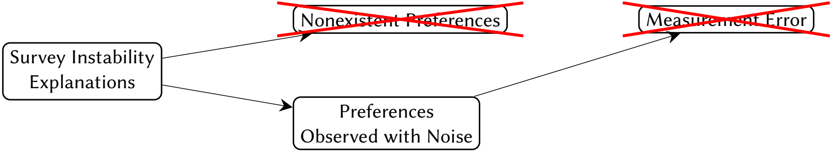

Survey Instability
Explanations
~~Nonexistent Preferences~~
Preferences
Observed with Noise
~~Measurement Error~~

What We Show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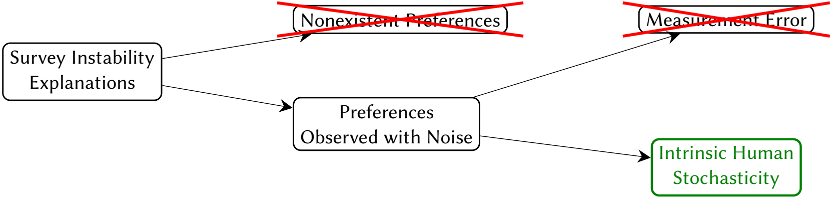
Survey Instability Explanations
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Preferences Observed with Noise
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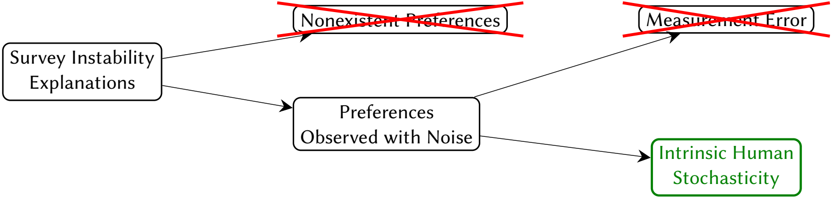
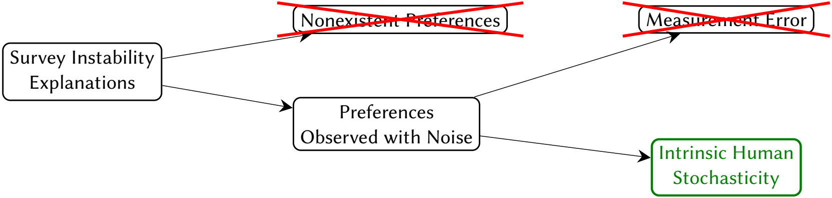
Intrinsic Human Stochasticity

What We Show



- Generally, in other fields: instrument v. subject variability

What We Show



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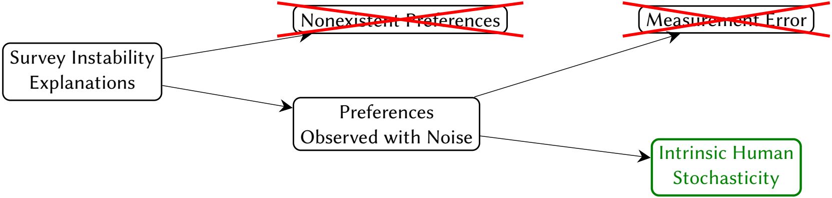
    - soft matter physics: tracking error v. thermal noise

What We Show



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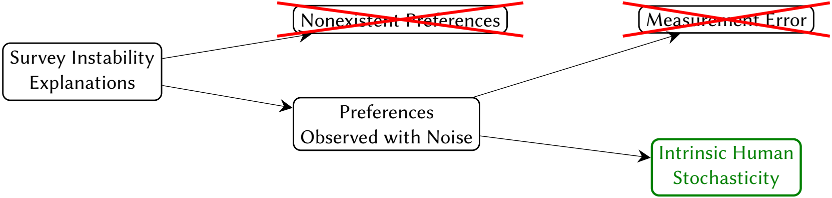
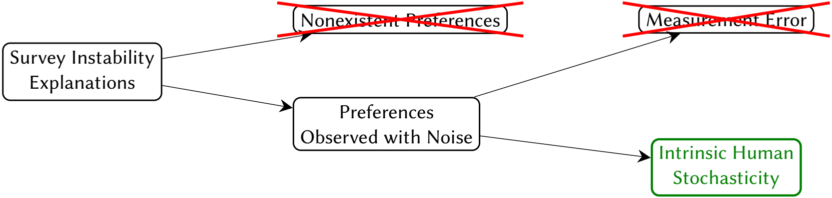
  - psychometrics: measurement error v. individual differences

What We Show



- Generally, in other fields: instrument v. subject variability
  - biology, medicine: technical v. biological variability

What We Show



- Generally, in other fields: instrument v. subject variability
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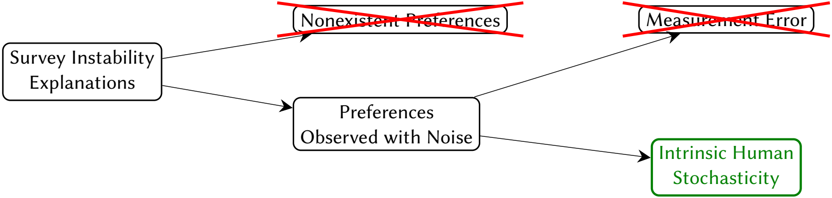

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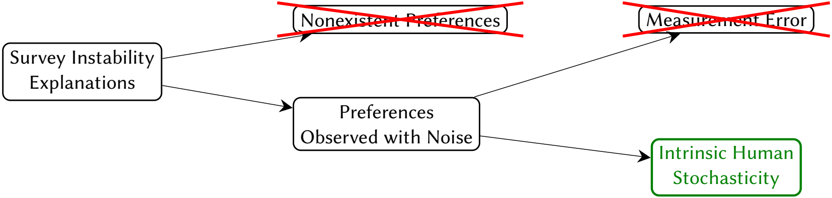

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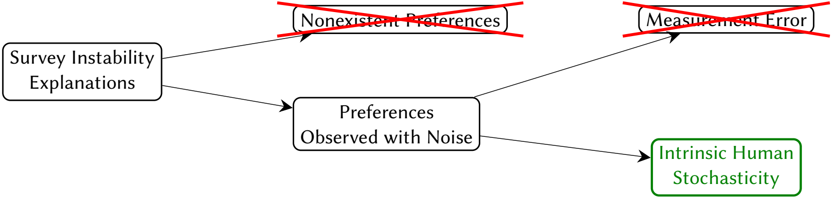

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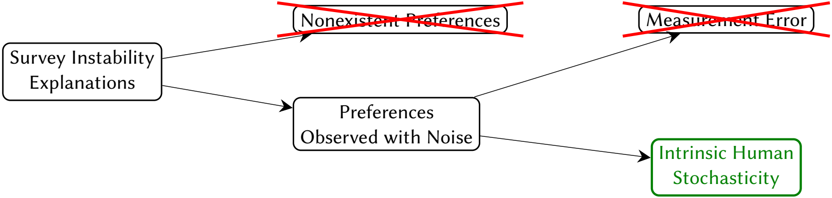

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- Imagine 2 CT Scans moments apart

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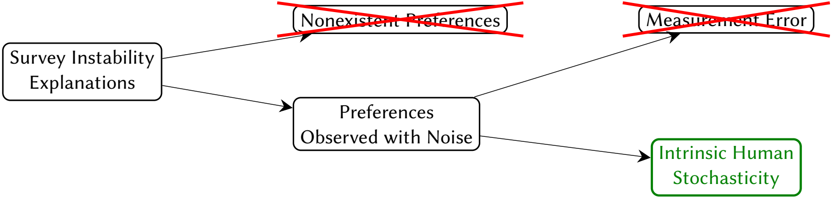

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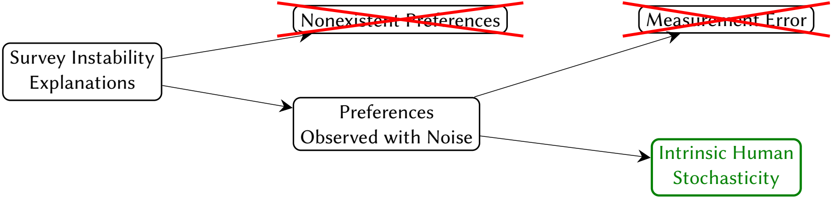

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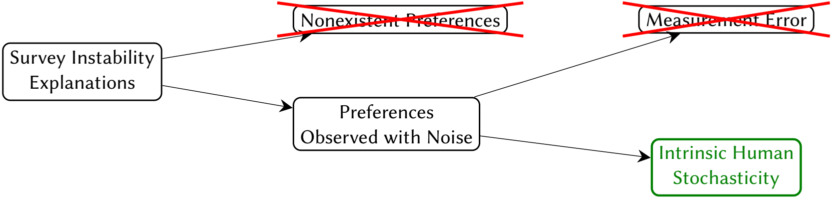

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- The creatures we study: variables, not constants

Sources of Stochasticity Leading to Survey Instability

Survey Instability

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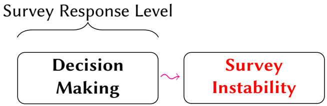

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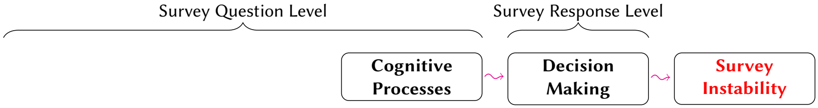

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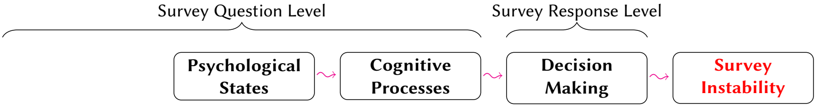

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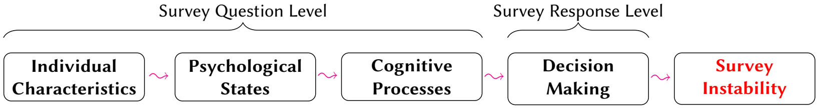

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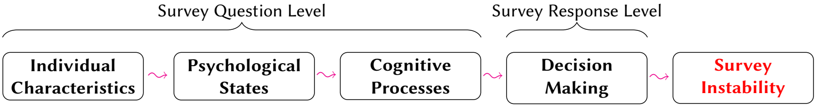


Used to find

Sources of Stochasticity Leading to Survey Instability

Survey Question Level

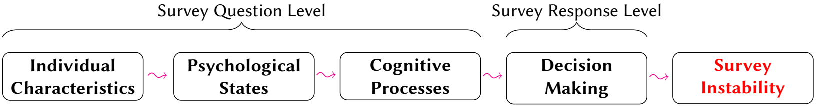
Survey Response Level

Individual Characteristics ⇝ **Psychological States** ⇝ **Cognitive Processes** ⇝ **Decision Making** ⇝ **Survey Instability**

Used to find

- Observable implications (of intrinsic human stochasticity)

Sources of Stochasticity Leading to Survey Instability



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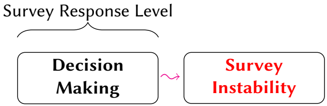

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Stochasticity Sources: Decision Making

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Election Day

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- 10 offices, each with Party A & B candidates

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- 10 offices, each with Party A & B candidates
- In 6/10, Party A is better for you than Party B.

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Election Day

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- In 6/10, Party A is better for you than Party B.
- How many Party A candidates will you choose?

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C_t

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$$C_t \left\{ \begin{array}{l} \\ \\ \end{array} \right.$$

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- Literature: Probability matching! (humans, bees, fish, pigeons...)

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- How Beliefs become Choices

$$C_t \begin{cases} = \mathbf{1}(\pi > 0.5) & \text{Expected Utility (deterministic)} & E(D) = 0 \\ = \mathbf{1}(\pi + \eta_t > 0.5) & \text{Random Utility (stochastic, } \eta_t \sim F) & E(D) > 0 \\ \sim \text{Bernoulli}(\pi) & \text{Probability Matching (stochastic)} & E(D) > 0 \end{cases}$$

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- Experiments:

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  - Result: still 40-60% stochastic!

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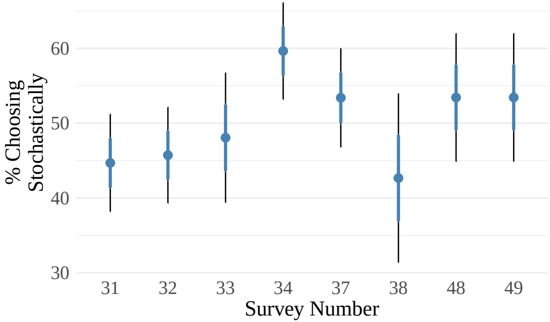
  - Result: still 40-60% stochastic! ☜

Stochasticity Sources: Decision Making

- Preference: $\rho \in \{0, 1\}$ (unknown)

Decision Making: Highly Stochastic!



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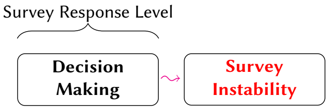
Stochasticity Sources: Cognitive Processing

Stochasticity Sources: Cognitive Processing

Survey Instability

Stochasticity Sources: Cognitive Processing



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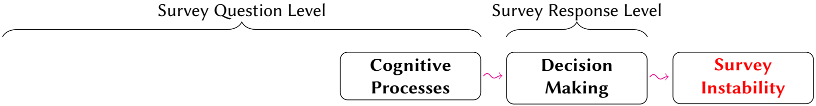

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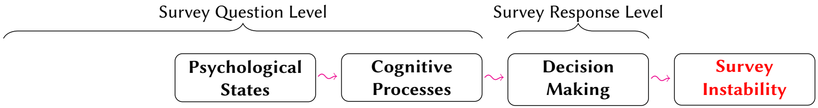
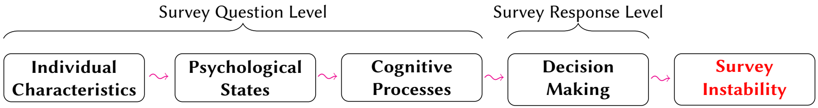


Stochasticity Sources: Cognitive Processing



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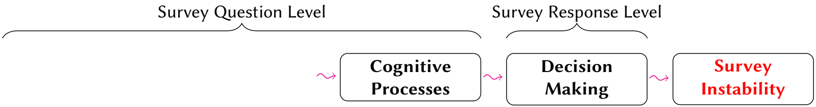

Stochasticity Sources: Cognitive Processing

What increases stochasticity (and thus instability)?

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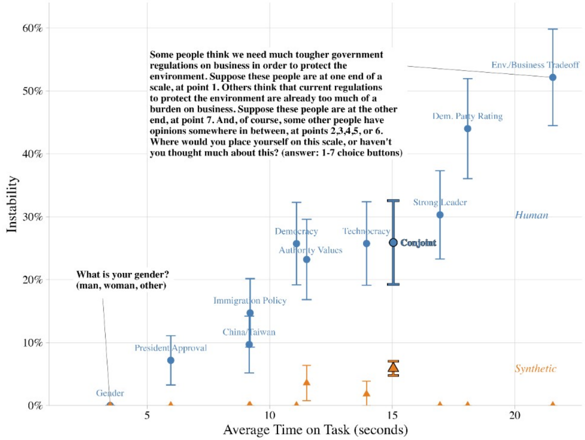
Stochasticity Sources: Cognitive Processing

What

- H
- L
- D



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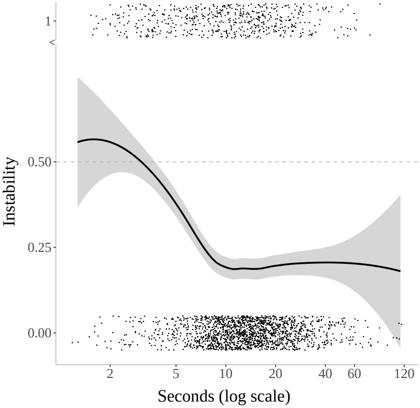
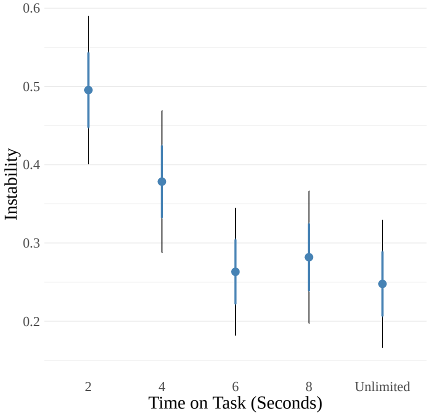
Stochasticity Sources: Cognitive Processing

Limited time-on-task causes survey instability



Observational



Experimental

Stochasticity Sources: Cognitive Processing

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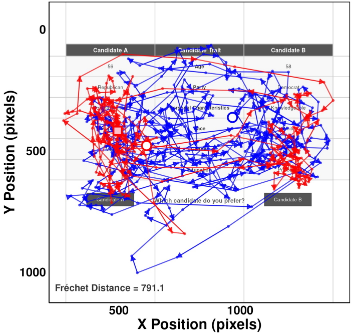
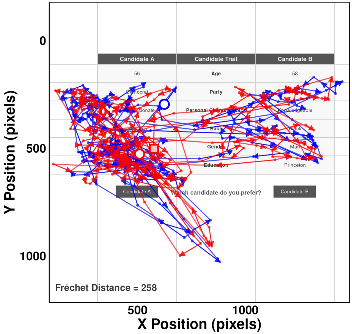
Stochasticity in Gaze Behavior



Large difference



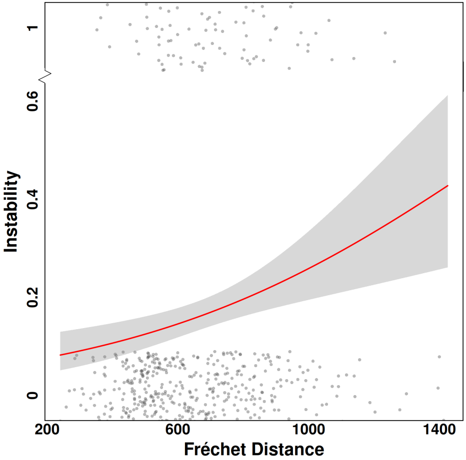
Small difference

Stochasticity Sources: Cognitive Processing

What

- H
- L
- D

Stochastic Distance ⇝ Survey Instability



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Stochasticity Sources: Psychological States

Survey Instability

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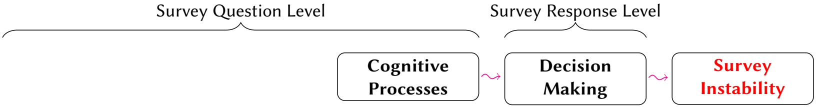
Survey Response Level

Decision Making $\rightsquigarrow$ **Survey Instability**

Stochasticity Sources: Psychological States



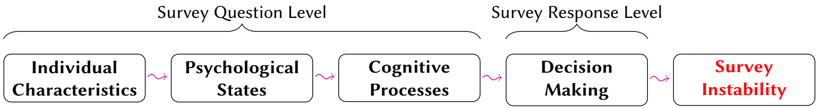
Stochasticity Sources: Psychological States

Survey Question Level: **Psychological States** ⇝ **Cognitive Processes**

Survey Response Level: **Decision Making**

Psychological States ⇝ **Cognitive Processes** ⇝ **Decision Making** ⇝ **Survey Instability**

Stochasticity Sources: Psychological States



Stochasticity Sources: Psychological States

Survey Question Level: **Individual Characteristics** ⇝ **Psychological States** ⇝ **Cognitive Processes**

Survey Response Level: ⇝ **Decision Making**

⇝ **Survey Instability**

Stochasticity Sources: Psychological States

Survey Question Level: **Individual Characteristics** ⇝ **Psychological States** ⇝ **Cognitive Processes**

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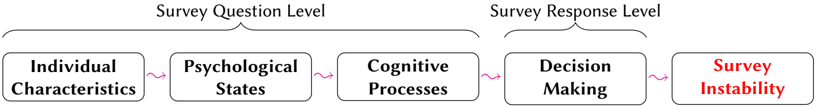
- Psychological States Summary

Stochasticity Sources: Psychological States



- Psychological States Summary
  - Attention Checks ruin surveys

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Survey Response Level: ⇝ **Decision Making** ⇝ **Survey Instability**

- Psychological States Summary
  - Attention Checks ruin surveys
  - Use Preoccupation instead

> Everyone comes to a new task (like this survey) with things on their minds. How preoccupied were you, just before you started this survey, with thoughts or worries about your partner, job, health, children, friends, money, or other concerns?
>
> {Very, Fairly, Slightly, Not at all} Preoccupied

So Who's to Blame for Survey Instability?

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Explanations:

So Who's to Blame for Survey Instability?

Explanations:

Survey Instability
~25%

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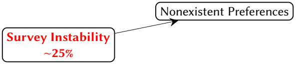
Explanations:



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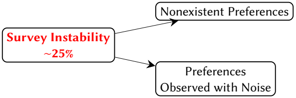

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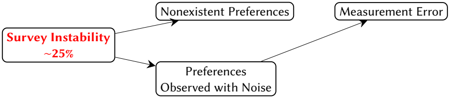

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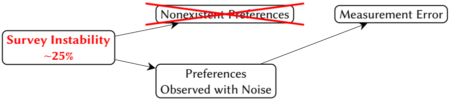

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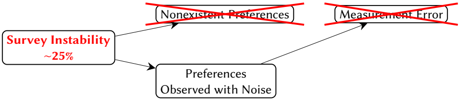

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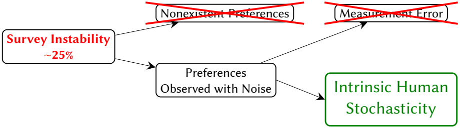

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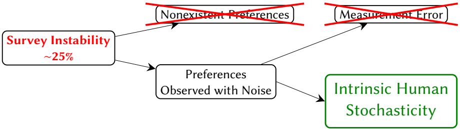

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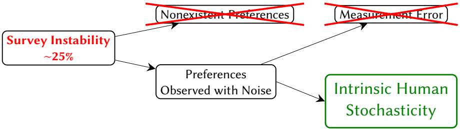

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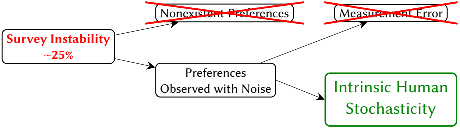

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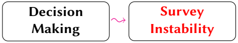

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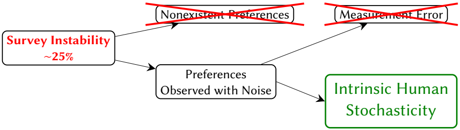

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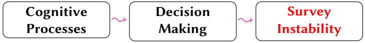

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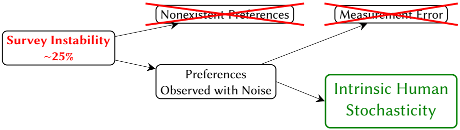

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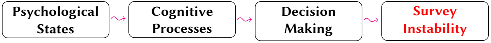

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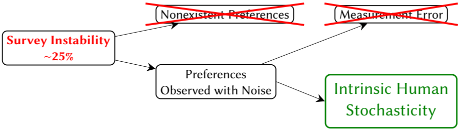

Sources:

Individual Characteristics ⇝ **Psychological States** ⇝ **Cognitive Processes** ⇝ **Decision Making** ⇝ **Survey Instability**

So Who's to Blame for Survey Instability?

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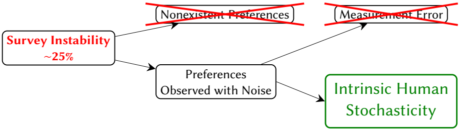

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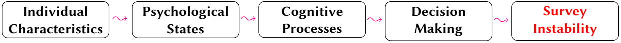

Implications:

So Who's to Blame for Survey Instability?

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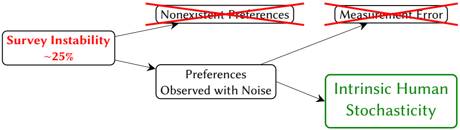

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Implications:

- Improve instruments:

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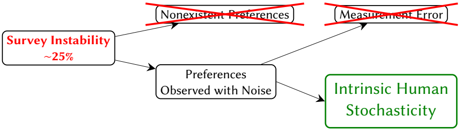

Sources:

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Implications:

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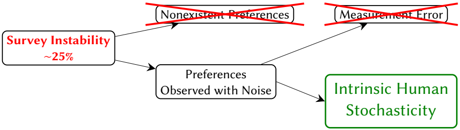

Sources:

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Implications:

- Improve instruments:
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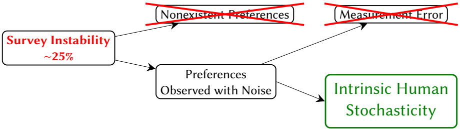
  - reduce cognitive complexity (simpler Qs, fewer As)

So Who's to Blame for Survey Instability?

Explanations:



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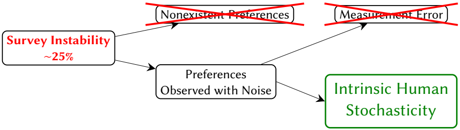

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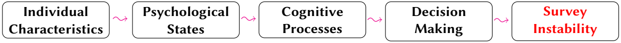

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- Filter likely-instable respondents:
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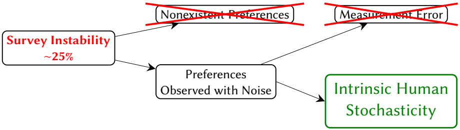
So Who’s to Blame for Survey Instability?

Explanations:



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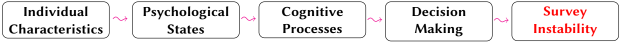

Implications:

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- Filter likely-instable respondents:
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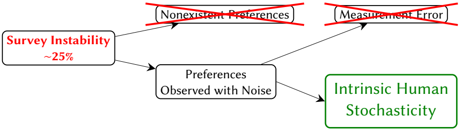
  - low time-on-task

So Who's to Blame for Survey Instability?

Explanations:



Sources:

Individual Characteristics ⇝ **Psychological States** ⇝ **Cognitive Processes** ⇝ **Decision Making** ⇝ **Survey Instability**

Implications:

- Improve instruments:
  - measure instability (one extra Q)
  - reduce cognitive complexity (simpler Qs, fewer As)
- Filter likely-instable respondents:
  - preoccupied (no attention checks)
  - low time-on-task
- Improve statistical methods:

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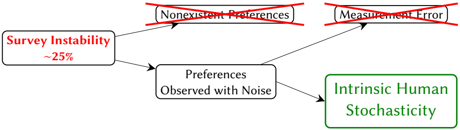

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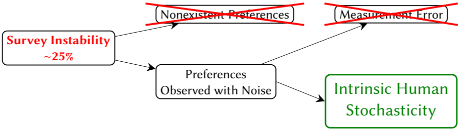

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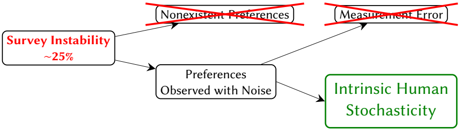

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Slides, data, paper: GaryKing.org